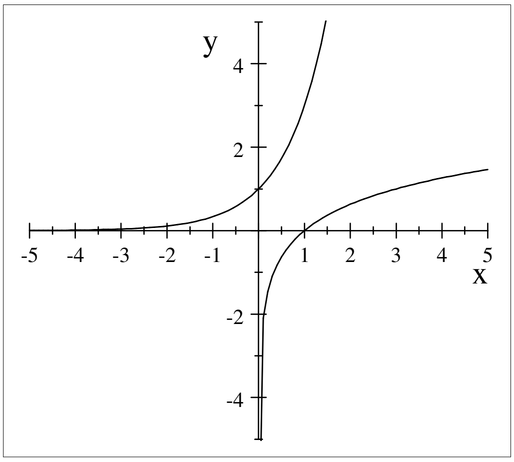
Worksheet on Logarithms

Definition

If $x = a^y$ (exponential form) then $y = \log_a x$ (logarithmic form) $a > 0; a \neq 1; x > 0$

Logarithmic laws

1. $\log_a PQ = \log_a P + \log_a Q \quad (a > 0; a \neq 1; P > 0; Q > 0)$
2. $\log_a \frac{P}{Q} = \log_a P - \log_a Q \quad (a > 0; a \neq 1; P > 0; Q > 0)$
3. $\log_a P^r = r \log_a P \quad (a > 0; a \neq 1; P > 0)$
4. $\log_a P = \frac{\log_s P}{\log_s a} + \log_a Q \quad (a, s > 0; a, s \neq 1; P > 0)$



Excercise 1

Simplify without the use of a calculator

1. $\log_2 8 + \log_3 27 - \log_7 7$
2. $2\log 60 - 2\log 2 - 2\log 3$
3. $5\log_4 2 - \log_4 0.125 - 2\log_4 8$
4. $2\log 5 - \log \sqrt[3]{64} + 4\log 2$

Excercise 2

Solve x without the use of a calculator

1. $\log x + \log(x-2) = 3\log 2$
2. $2\log_2 5 - \log_2(x-1) = 1$
3. $2\log_5 x - 3 = -\frac{1}{\log_5 x}$

Logarithmic inequalities

$\log_a x < b; x > 0; a > 0; a \neq 1$

If $a > 1$ (increasing function) then $x < a^b$ (inequality remains the same).

Excercise 3

Sketch the graph and solve the inequality

1. $\log_3 x < 2$

Excercise 4

Solve the inequality

1. $\log_2(3x-2) + \log_2(x+1) \leq 3$